

Syllabus for

Academic year 2021/2022 >

MVE030 - Fourier analysis

Fourieranalys

Syllabus adopted 2022-05-02 by Head of Programme (or corresponding)

Owner: [TKTFY](#)

6,0 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: First-cycle

Main field of study: Chemical Engineering with Engineering Physics, Mathematics, Engineering Physics

Department: 11 - MATHEMATICAL SCIENCES

Teaching language: Swedish

Application code: 57142

Open for exchange students: No

Maximum participants: 180

Only students with the course round in the programme plan

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course 	No Sp	Examination dates 
0105 Examination 6,0 c Grading: TH							Fri 18/03-2022 am J DIG  Wed 08/06-2022 am J DIG  Tue 23/08-2022 am J DIG

In programs

[TKTEM ENGINEERING MATHEMATICS, Year 2 \(compulsory\)](#)

[TKTFY ENGINEERING PHYSICS, Year 2 \(compulsory\)](#)

Examiner:

[Julie Rowlett](#)

 [Go to course homepage](#)

Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.

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Course specific prerequisites

Real analysis, Multivariable analysis, Linear algebra, Complex mathematical analysis.

Aim

The course introduces Fourier methods in the program. These methods are powerful mathematical tools in technology and science.

Learning outcomes (after completion of the course the student should be able to)

After completing the course, the student will be able to solve partial differential equations using separation of variables, eigenfunction and Fourier series expansions, eigenfunction expansions using Sturm-Liouville problems, as well as Fourier and Laplace transforms. In addition, the student will be able to apply the theoretical concepts of Hilbert spaces to solving physical problems. In this regard, the student will be able to determine, based on the geometry of the physical domain and the character of the equation, which orthogonal system, such as trigonometric functions, Bessel functions, or orthogonal polynomials, is best suited to solve the physical problem. The student will also be able to use Fourier series to compute sums, like the sum of $1/n^2$. The student will learn how to use Fourier transforms to compute tricky integrals.

Content

The method of separation of variables. Trigonometric Fourier series and their convergence. Examples of initial and boundary value problems for partial differential equations: the heat equation, the wave equation, Laplace/Poisson's

equation. Orthogonal systems of functions, completeness, Sturm-Liouville eigenvalue problems. Various solution techniques like homogenization, superposition and eigenfunction expansion. Bessel functions and orthogonal polynomials (Legendre, Hermite and Laguerre polynomials). Solution methods in spherical or cylindrical coordinates. Fourier transforms and their applications to partial differential equations. Signal analysis, discrete Fourier transforms and Fast Fourier transforms. Laplace transforms with applications.

Organisation

The course is organized in lectures and exercises (about 5h/week of each). Computerized hand-in assignments may occur. The course and the examination coincide for the programs F and TM.

Literature

Custom course book plus some additional material.

Examination including compulsory elements

Written exam with about 6 problems and 2 theory questions (5 hours).

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.